Intak Hwang

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I'm a Ph.D. student in Cryptography & Privacy Lab at Seoul National University, advised by Yongsoo Song. I'm interested in post-quantum cryptographic protocols based on lattices, including but not limited to Fully Homomorphic Encryption and Zero-Knowledge Proofs.

Education

Seoul National University 2023 — Present
Integrated M.S. & Ph.D. in Computer Science & Engineering
Advised by Yongsoo Song

DGIST 2018 — 2022
B.S. in School of Undergraduate Studies *Summa Cum Laude*

Conference Publications

[C10] 2025/395
On the (In)security of Approximate Computation Protocols from CKKS
with Yisol Hwang, Miran Kim, Dongwon Lee, Yongsoo Song
Asiacrypt 2026

[C09] 2025/1804
HERDS: Multi-Key Fully Homomorphic Encryption with Sublinear Bootstrapping
with Binwu Xiang, Seonhong Min, Zhiwei Wang, Haoqi He, Yuanju Wei, Kang Yang, Jiang Zhang, Yi Deng, Yu Yu
Eurocrypt 2026

[C08] 2024/2032
Carousel: Fully Homomorphic Encryption with Bootstrapping over Automorphism Group
with Seonhong Min, Yongsoo Song
Asiacrypt 2025

[C07] 2025/382
On the Security and Privacy of CKKS-based Homomorphic Evaluation Protocols
with Seonhong Min, Jinyeong Seo, Yongsoo Song
Asiacrypt 2025

[C06] 2025/216
Practical TFHE Ciphertext Sanitization for Oblivious Circuit Evaluation
with Jinyeong Seo, Seonhong Min, Yongsoo Song
ACM CCS 2025

[C05] 2024/1879

Practical Zero-Knowledge PIOP for Maliciously Secure Multiparty Homomorphic Encryption

with Hyeonbum Lee, Jinyeong Seo, Yongsoo Song

ACM CCS 2025

[C04] 2025/1255

Efficient Full Domain Functional Bootstrapping from Recursive LUT Decomposition

with Shinwon Lee, Seonhong Min, Yongsoo Song

SAC 2025

[C03] 2024/1502

MatriGear: Accelerating Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing

with Hyunho Cha, Seonhong Min, Jinyeong Seo, Yongsoo Song

IEEE S&P 2025

[C02] 2024/306

Concretely Efficient Lattice-based Polynomial Commitment from Standard Assumptions

with Jinyeong Seo, Yongsoo Song

Crypto 2024

[C01] 2023/1328

Optimizing HE Operations via Level-aware Key-switching Framework

with Jinyeong Seo, Yongsoo Song

WAHC 2023

Journal Publications

[J01]

A Privacy-Preserving HLA Imputation Method with Homomorphic Encryption

with Hakin Kim, Yongsoo Song, Buhm Han

iScience

Preprints

[P04] 2026/2809

Pastel: Single Secret Leader Election with Private Weights

with Jung Hee Cheon, Kevin Choi, Jaehee Kang, Alireza Kavousi, Junho Lee, Yongdong Yeo

[P03] 2026/2120

Multi-Key FHE with Compact Ciphertexts (or: MKFHE Strikes Back)

with Seonhong Min, Yongsoo Song

[P02] 2026/044

Jindo: Practical Lattice-Based Polynomial Commitments for Client-Side Proving

with Hyeonbum Lee, Jinyeong Seo, Yongsoo Song

[P01] 2025/203

Ciphertext-Simulatable HE from BFV with Randomized Evaluation

with Seonhong Min, Yongsoo Song

Presentations

Practical Zero-Knowledge PIOP for Maliciously Secure Multiparty Homomorphic Encryption

FHE.org 2026 — Taipei, Taiwan

Practical TFHE Ciphertext Sanitization for Oblivious Circuit Evaluation

ACM CCS 2025 — Taipei, Taiwan

FHE.org — Online

MatriGear: Accelerating Authenticated Matrix Triple Generation with Scalable Prime Fields via Optimized HE Packing

IEEE S&P 2025 — San Francisco, USA

Optimizing HE via Level-aware Key-switching

WAHC 2023 — Copenhagen, Denmark

Projects

TFHE-go

TFHE-go is an implementation of the (MK)TFHE scheme, written in Go and Go assembly. It is currently one of the fastest and most feature-complete open-source TFHE implementations available.

Ringo-SNARK

Ringo-SNARK is a Zero-Knowledge PIOP toolkit for efficiently proving Ring-LWE relations, written in Go. It supports simple, gnark-like circuit design and compilation.

Honors and Scholarships

National Cryptographic Contest

Excellence Award, Encouragement Award	2025
Best Award, Excellence Award	2024
Special Award	2023

CTF Security Competitions	2020 — 2022
SSTF Hacker's Playground 2022	*5th place*
WhiteHat Contest 2021	*3rd place*

DEFCON CTF 2021	*Finalist*
PlaidCTF 2021	*5th place*
Real World CTF 2020/2021	*1st place*
Midnight Sun CTF 2020 Finals	*7th place*
TokyoWesterns CTF 2020 Finals	*3rd place*
DEFCON CTF 2020	*Finalist*

DGIST Dean's List 2020

Academic Services

Program Committee
FHE.org 2027

External Reviewer
Asiacrypt 2026, WAHC 2026, PKC 2027, Eurocrypt 2027

Skills

Languages
Korean (native), English (fluent)

Programming Languages
Go, Python (SageMath), C/C++, Rust, LaTeX

Other Activities

Member of CTF Team CodeRed 2020 — 2022
I participated in CTF competitions from time to time, mostly solving crypto challenges.